CAPSTONE PROJECT SUBMISSION

AI/ML Fundamentals Course — Module 8

Full Name

Mubinabegim To'lqinjonova

Project Track

Field-Based Scenario Track — EDU-02: Student Performance Early-Warning Model (EdTech)

Project Title

Student Performance Early-Warning Model — predicting academic risk from early-course data (OULAD)

Repository URL

https://github.com/Mubina-lazy/edu02-early-warning-model

Short Project Description

A binary classification system that flags students at risk of failing or withdrawing from an online course, using only information available in the first ~25% of the course: demographics, virtual learning environment (VLE) click activity, and already-due early assignment scores. Built on the Open University Learning Analytics Dataset (OULAD, CC-BY 4.0; 32,593 students, 22 course presentations).

The data pipeline is leakage-safe by construction: a documented data audit with an issue log, a chronological split by course presentation (train on 2013 cohorts, validate on 2014B, test once on the unseen 2014J cohort), and explicit bans on any post-cutoff information. Seven experiment runs (DummyClassifier and Logistic Regression baselines, weighted Logistic Regression, two Random Forests, two XGBoost variants) were tracked in MLflow; the final model is XGBoost with a decision threshold of 0.327 tuned on validation.

Final one-shot test results (2014J): recall 0.760 / precision 0.583 / F1 0.660 / PR-AUC 0.714 for the at-risk class, versus zero recall for the majority baseline. The report includes per-module results (including a documented cold-start course), fairness slices (no recall gap for students with disabilities), and a false-negative error analysis. The repository is fully reproducible from raw data with pinned dependencies.

Demo / Application Link

Reproducible Google Colab demo: demo.ipynb in the repository root (open in Colab and Run all; no dataset download required — the trained model artifact ships with the repository).

Access Instructions

The repository is public — no special access is required. All code, evaluation reports, the trained model artifact, and the Colab demo are available directly from the link above.

AI assistance disclosure: the project was developed with the help of an AI coding assistant (Claude) under the course rules for Module 8, with all decisions reviewed, tested, and understood by the student; assistance is visible in the repository's commit history.